

PLC Principles of Operation

1 Control Problem: Mixer Motor Automation

Stirring Application

Objective: A mixer motor must automatically stir the liquid in a vat based on specific process conditions, while also allowing for manual control.

- **Automatic Condition:** The motor should run when *both* the temperature and pressure reach preset values.
- **Manual Condition:** Direct operation is provided via a separate pushbutton station.
- **Monitoring:** Temperature and pressure sensor switches close their contacts when preset values are reached.

2 Hardware Configuration

2.1 Input Field Devices

These devices are hardwired to the PLC's input module. In this scenario, we use a **120 VAC modular configuration**.

- **Pressure Switch:** Closes upon reaching preset pressure.
- **Temperature Switch:** Closes upon reaching preset temperature.
- **Pushbutton:** Provides manual start capability.

2.2 Output Field Devices

These are the physical actuators controlled by the PLC.

- **Motor Starter Coil:** Hardwired to a 120 VAC output module. When energized, it engages the mixer motor.

Addressing Location Scheme

Every field device is assigned a specific **address** according to the manufacturer's scheme. This address allows the PLC to identify the exact physical terminal where each device is connected.

3 PLC Logic and Operating Cycle

The Ladder Logic Program

The control logic is constructed as a **ladder logic program** using a programming device and then transferred into the **CPU user memory**.

- **Symbols:** Represent logical instructions (contacts, coils).
- **Numbers:** Represent the physical I/O addresses of the hardware devices.

3.1 RUN Mode Operation Scheme

Logic Continuity & Memory Status

When a PLC is in **RUN Mode**, the processor evaluates the ladder program and records the status of field devices in the controller's memory:

- **Logic 1 (TRUE):** Represents a **closed contact**.
- **Logic 0 (FALSE):** Represents an **open contact**.

Execution Flow:

- The CPU evaluates each internal contact's status.
- If the states of the input contacts provide **logic continuity** from left to right on a rung:
 - Output coil memory location is set to **Logic 1**.
 - Output module interface contacts are **closed**.
- If there is **no logic continuity**:
 - Output coil memory location remains **Logic 0**.
 - Output module interface contacts stay **open**.

3.2 Scan and Timing

The PLC Scan Cycle

A **Scan** is the completion of one full cycle of the input-execute-output sequence.

- **Scan Time:** The time required for one full cycle. This determines the **speed of response** of the PLC.
- **Internal Update:** During the program scan, only the **output memory location** is updated.
- **I/O Scan:** The **actual physical output** is updated after the program scan completes (Integrated I/O update).

4 Wiring and Installation

Installation Steps for Mixer Process

Setting up the physical control circuit involves hardwiring the field devices according to the terminal assignments:

1. **Power Connection:** Fused power lines are connected to the controller's **L1** and **L2** terminals.
2. **Input Hardwiring:** Field input devices (switches, buttons) are connected between **L1** and the respective input terminals (e.g., **I1**, **I2**, **I3**).
3. **Output Hardwiring:** The motor starter coil is connected:
 - Directly to the common **L2** line.
 - In series with the **Q1** relay output contacts to the **L1** line.
4. **Programming:** The ladder logic is entered using the front keypad/LCD display or via ****Programming Software**** on a PC.

5 Advanced Case Study: Modified Process Control

Modified Logic Requirements

The process requirements are updated for a new operation mode:

- **Condition:** The manual pushbutton should now be able to operate the mixer at **any pressure** level.
- **Constraint:** It must still respect the specified **temperature setting**.

Ladder Logic Change

In the revised program, the pushbutton contact is moved to be in **parallel with the pressure switch only**, while the temperature switch contact remains in series with the resulting parallel branch.

Accurate hardwiring combined with flexible logic definitions allows for complex industrial control adaptations.